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RESEARCH ARTICLE

Predicting invasion success of naturalized cultivated plants in China

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Abstract

1. Plant invasions pose significant threats to native ecosystems, human health, and global economies. However, the complex and multidimensional nature of factors influencing plant invasions makes it challenging to predict and interpret their invasion success accurately.
2. Using a robust machine learning algorithm, random forest, and an extensive suite of characteristics related to environmental niches, species traits, and propagule pressure, we developed a classification model to predict the invasion success of naturalized cultivated plants in China. Based on the final optimal model, we evaluated the relative importance of individual and grouped variables and their prediction performance.
3. Our study identified key individual variables within each of three groupings: climatic suitability and native range size (environmental niches), phylogenetic distance to the closest native taxon and vegetative propagation mode (species traits), and the number of botanical gardens and provinces where species were cultivated (propagule pressure). Remarkably, when grouped variables were evaluated, the relative importance of grouped variables increased dramatically—by 13.5–17.7 times—compared with the cumulative importance of individual variables within a category. However, the relative importance of one category was primarily due to the number of variables within each category rather than its inherent characteristics.
4. *Synthesis and applications*. Our findings emphasize the necessity of developing data-driven predictive tools for effective invasion risk assessment using large datasets. By identifying key individual variables, we recommend prioritizing surveillance of alien plant species with large native ranges and high climatic suitability. By evaluating grouped variables, we emphasize the significance of grouped variables in enhancing model interpretability by providing deeper insights into the complex interactions among individual factors within each predefined category. This comprehensive approach can not only identify the most influential predictors of invasion success but also equip policymakers with evidence-based strategies for surveillance, early detection, and targeted intervention of invasive plants.

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KEYWORDS
environmental niches, feature importance, individual and grouped variables, naturalized cultivated plants, plant invasions, propagule pressure, random forest, species traits

1 | INTRODUCTION

Plant invasions increasingly threaten native biodiversity, ecosystem functions, human health, and economies on a global scale (Pyšek et al., 2020; Vilà & Hulme, 2017). Around 3000 vascular plant species have been documented as being invasive outside their native ranges globally (Laginhas et al., 2023), representing roughly one-quarter of all known naturalized vascular plant species (van Kleunen et al., 2015). The global accumulation of established alien plants is still increasing without saturation (Seebens et al., 2017). Therefore, it is necessary for ecologists to elucidate the mechanisms that enable some alien plants to become invasive while others do not (van Kleunen, Dawson, et al., 2010). As most alien plants that have become invasive were intentionally introduced for cultivation (van Kleunen et al., 2018; Yan et al., 2019), analysing the inventories of cultivated alien plants presents a unique opportunity to fill these knowledge gaps.

Research on plant invasiveness has primarily focused on species traits. Species traits related to resource capture and competition (e.g., plant height), longevity (e.g., lifeform), and offspring recruitment (e.g., propagation mode) are often linked to invasion success (Daehler, 2003; Pyšek & Richardson, 2008; van Kleunen, Weber, & Fischer, 2010). In general, tall-statured plant species have a greater capacity to accumulate biomass and grow more rapidly than short-statured ones (Canavan et al., 2019). Plant species with short lifespans (annuals or short-lived perennials) and high seed production can rapidly expand their populations and colonize new areas (Beckman et al., 2018; Mologni et al., 2022). These species traits offer invasive plants advantages over native species, allowing them to establish, spread, and persist in new environments (Gioria et al., 2023).

Environmental niches, that is, the range of environmental conditions in which a species can survive and reproduce, also influence invasion success (Petitpierre et al., 2016; Theoharides & Dukes, 2007). For instance, many invasive plants exhibit climatic niche conservatism, meaning that they occupy similar environmental conditions in their introduced range as they do in their native range (Liu et al., 2020; Petitpierre et al., 2012). However, the extent to which invasive plants exhibit climatic niche conservatism remains an ongoing debate as certain invasive plants undergo niche shifts in their introduced ranges (Atwater et al., 2017; Bates & Bertelsmeier, 2021). It is essential to determine whether environmental niches can be a reliable predictor of invasion success for most alien plants.

Propagule pressure, which refers to the number of individuals introduced and the frequency of introduction events, plays a significant role in the introduction and establishment of alien plants (Lockwood et al., 2005). The collection and cultivation of living plants via botanical gardens not only facilitates the establishment of alien plants in introduced ranges but also accelerates their escape into natural habitats (Hulme, 2011). Moreover, the selection of alien plants is often intentional, prioritizing species with specific traits, such as fast growth (Chrobock et al., 2011) or cold hardiness (Maurel et al., 2016) and those with economic uses (van Kleunen et al., 2018, 2020). This non-random selection may inadvertently lead to the introduction of alien plants with a higher potential for naturalization and invasiveness. Accurate prediction of invasion success thus requires considering propagule pressure, including factors such as planting frequency and the economic use status of alien plants. Overall, these three categories—environmental niches, species traits, and propagule pressure—form the foundation for understanding plant invasion mechanisms.

Random forest (RF), a robust machine learning (ML) algorithm, is increasingly used in invasion ecology studies for tasks such as classifying species and predicting invasion risk (Chen et al., 2015; Cutler et al., 2007; Keller et al., 2011). RF offers inherent advantages over many other ML algorithms by handling nonlinear relationships and variable interactions without prior specification, managing missing data, and balancing the bias-variance trade-off (Breiman, 2001). Despite these advantages, applying RF in invasion ecology still faces challenges. For instance, although RF can handle large datasets, ecological data for classifying invasive species is often limited, with studies frequently constrained by small sample sizes of alien species (e.g., Chen et al., 2015; Keller et al., 2011). In addition, while RF effectively captures complex ecological relationships, the increased model complexity from incorporating additional features may impede the derivation of actionable insights and elucidation of underlying ecological mechanisms (Pichler & Hartig, 2023). Given that the factors determining invasion success are multidimensional and that large open-access ecological datasets are readily available, it is worthwhile to investigate whether RF remains a promising tool for leveraging these extensive datasets (Novoa et al., 2020).

In this study, we examined all naturalized cultivated plants in China by analysing their characteristics related to species traits, environmental niches, and propagule pressure to identify factors influencing invasion success. We also considered the phylogenetic relationships between the alien and native floras to account for the potential influence of evolutionary history and the similarity of species. We used RF to evaluate the relative importance of the 34 individual variables and the relative importance of the three categories. Specifically, we addressed three key questions: (1) Is RF an efficient tool for predicting invasion success? (2) Which individual variable is most important for predicting invasion success? (3) Which grouped variable is most effective in predicting invasion success? By addressing these questions, we seek to deepen the understanding of plant invasions and increase the capacity to predict and manage invasive species in China.

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2 | MATERIALS AND METHODS

2.1 | Data compilation

We compiled a checklist of the 735 naturalized plant taxa introduced for cultivation in China, based on the *Catalogue of Cultivated Plants in China* (Lin, 2018) and *The Checklist of the Naturalized Plants in China* (Yan et al., 2019). The binomial names of these naturalized taxa were standardized according to The Plant List (TPL, version 1.1; http://www.theplantlist.org) using the R package 'Taxonstand' (Cayuela et al., 2021). Among these naturalized taxa, 435 were classified as non-invasive and 300 as invasive in China, based on information from Hao and Ma (2023). Non-invasive taxa are naturalized taxa that form self-sustaining populations outside cultivation but remain limited in spread. Invasive taxa are naturalized taxa that spread widely, causing economic, social, and ecological damage to the invaded ecosystems. These definitions are consistent with those used by Lin et al. (2021).

To identify the characteristics that could potentially drive the invasion success of naturalized taxa, we compiled data on 12 characteristics for each taxon, resulting in 34 variables. The 34 variables were grouped into three categories, including species traits and similarity to native species (lifeform, propagation mode, maximum height, phylogenetic mean pairwise distance [MPD], phylogenetic nearest pairwise distance [NNPD], and weighted mean pairwise distance [wMPD] to the native flora in China), propagule pressure (the number of botanical gardens, the number of provinces in which taxa were cultivated, economic use category and the number of economic use categories), and environmental niches (climatic suitability, native range size, and continents of origin). Native range size and continents of origin were quantified using the Taxonomic Databases Working Group (TDWG) level-1 (continental) and level-3 (regional) geographical classifications. A phylogenetic tree was constructed using 735 naturalized plant taxa and 30,248 native plant taxa in China. Further details are provided in the Supporting Information.

2.2 | Data preparation

There were 38 naturalized taxa with missing values for some of the 34 variables. Following the guidance of Breiman (2003), missing data were imputed by weighting the frequency of the non-missing values with proximity values, using the 'rfImpute' function in the R package 'randomForest' (Feng et al., 2020).

To standardize the explanatory variables, we scaled all continuous variables to have a mean of zero and a standard deviation of one (Table S1). Before scaling, we also natural $\log(x+1)$ transformed the number of botanical gardens, the number of provinces, the number of economic use categories, and native range size; natural $\log(x+0.001)$ transformed climatic suitability; natural log-transformed MPD, NNPD, wMPD, and maximum height to achieve more regular distributions of these variables. As maximum height is associated with the lifeform, we natural log-transformed and scaled maximum height separately within each life-form category.

To preliminarily explore the rationality of variable grouping, we employed Pearson correlation analysis to examine bivariate relationships and self-organizing map (SOM) analysis to visualize high-dimensional data clustering patterns through dimensionality reduction (Kohonen, 2001). Further details are provided in the Supporting Information.

2.3 | Random forest with class weights

We randomly divided the dataset, stratified by class (i.e. the invasive taxa as the positive class, and non-invasive taxa as the negative class), into training (80%; 588 taxa) and test (20%; 147 taxa) sets. The training set was used for model fitting, and the test set was used to evaluate model performance and avoid unrealistically low error rates returned by in-sample comparisons (Schmidt et al., 2012).

To mitigate the risk of bias towards the majority class due to imbalanced data, we followed the idea of cost-sensitive learning and incorporated class weights into the RF for the training set, thereby penalizing misclassifications of the minority class (Chen et al., 2004; Valavi et al., 2021). To obtain the optimal RF, we conducted a grid search to optimize two hyperparameters: *mtry* (the number of randomly selected explanatory variables) and *splitrule* (the splitting rule). RF was implemented using the 'ranger' model in the R package 'caret' (Kuhn, 2022).

Partial dependence plots were constructed to characterize relationships between individual variables and the invasion probabilities predicted by the optimal RF, using the 'model_profile' function in the R package 'DALEX' (Biecek, 2018).

The permutation-based relative importance of individual and grouped variables in the optimal RF was estimated, using the 'feature_importance' function in the R package 'ingredients' (Biecek & Baniecki, 2023). The permutation importance of one individual variable or category was estimated by calculating the error increase by one minus the area under the ROC curve after the values of one variable or all variables within one category were permuted. This process was repeated 1000 times. Larger increases in the error values indicate more important variables or categories in the model. The proportion of relative importance was calculated by separately calculating the proportion of the mean error increase of one variable or category to the total error increase caused by all variables across the categories. Moreover, we conducted a randomization test to examine whether the relative importance of the category correlated with the number of individual variables within each category.

The predictive performance of the optimal RF on the test set was evaluated using six metrics: accuracy, sensitivity, specificity, F1-score, *G*-mean, and AUC.

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The model adequacy was determined by comparing its accuracy to the no-information rate. McNemar's test examined the difference between errors in the positive and negative predictions of the optimal RF (Kuhn & Johnson, 2013). These statistical tests were implemented using the 'confusionMatrix' function in the R package 'caret' (Kuhn, 2022).

3 | RESULTS

3.1 | The roles of single variables in invasion success

Of the 34 variables, 28 contributed significantly to the RF in predicting the invasion success of naturalized cultivated plants in China (Figure 1; Figure S2). Regarding environmental niche variables, naturalized taxa originating from over 17 TDWG level-3 regions and/or exhibiting climatic suitabilities of 26% and higher in China were more likely to become invasive (Figure 1a,b; Figure S1A,B). Besides, taxa from Northern and/or Southern America had a higher likelihood of being invasive than those from other continents (Figure 1i,j; Figure S1I,J).

Propagule pressure variables were also important for predicting invasion success. Naturalized taxa with one or more economic uses and cultivated in intermediate botanical gardens and provinces had a higher likelihood of invasion than those without these characteristics (Figure 1l,n; Figure S1L–N). Some economic uses increased the likelihood of invasion, while others did the opposite (Figure 1o–x). Taxa used for animal food, medicines, and poisons were more likely to become invasive (Figure 1o,v,w). Conversely, taxa not used for environmental uses, fuels, gene sources, human food, and materials had a higher likelihood of invasion (Figure 1p–s,u). Taxa used for invertebrate food and social uses showed no significant difference (Figure 1t,x).

Finally, among species trait variables, naturalized taxa that were short-lived, seed-propagated, and tall-statured, were more likely to become invasive (Figure 1y–ae; Figure S1Y–AE). Besides, three phylogenetic metrics consistently indicated that naturalized taxa more closely related to the native flora were more likely to become invasive (Figure 1af–ah; Figure S1AF–AH).

3.2 | Relative importance of single and grouped variables

When assessing individual variables, they collectively accounted for only 6.17% of the total model predictability (Figure 2; Figure S2). For environmental niches, climatic suitability, native range size, and one native continent (Southern America) were identified as the three most important individual variables. For species traits, two phylogenetic metrics (NNPD and wMPD) and one propagation mode (vegetative) emerged as the three most important variables. For propagule pressure, the number of provinces and botanical gardens and one type of economic use (materials) were the three most important variables.

When assessing three categories, these categories collectively accounted for 98.9% of the total model predictability (Figure 3; Figure S3). The three categories were effectively equally important, explaining 32%–34% of the total model predictability (Figure 3). Meanwhile, the relative importance of each category was not significantly greater than expected from the random assignment of variables to each category (Figure S4).

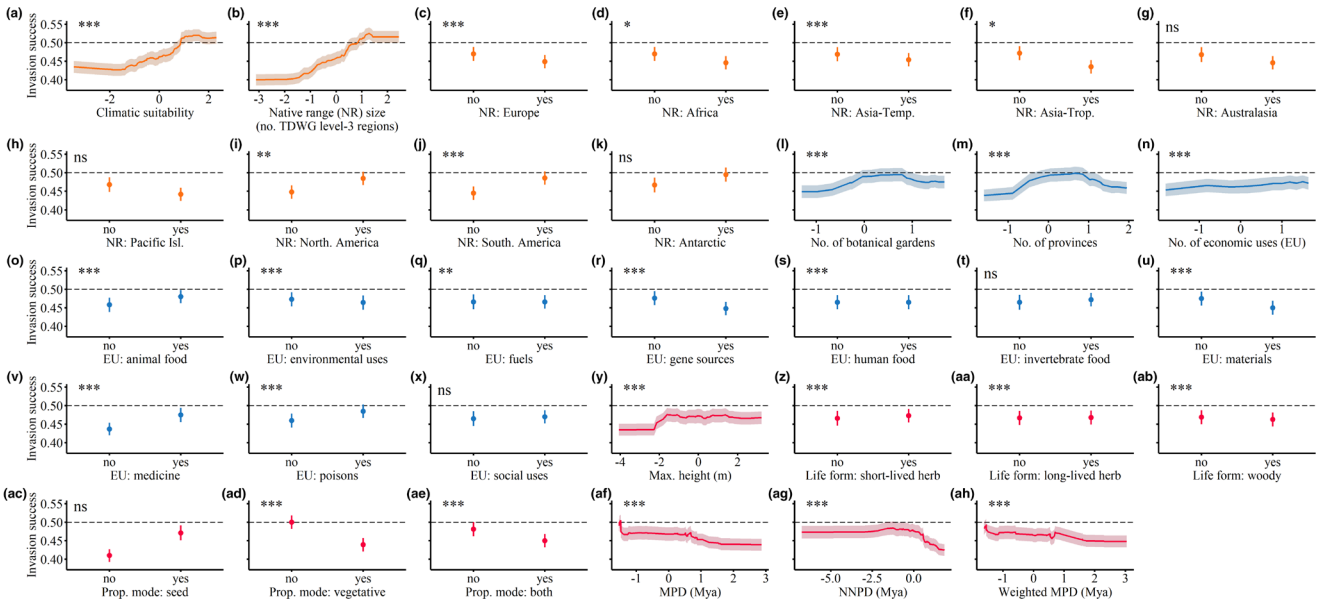

FIGURE 1 Partial dependence plots (means ± 95% CI) for each variable in the RF. The scaled data, on which the analyses are based, are shown, and the raw data are shown in Figure S1. The dashed line indicates a 50% probability of invasion success. The colours used for the different variables correspond to the categories they belong to (orange: Environmental niches, red: Species traits, blue: Propagule pressure). EU, economic use; NR, native range. The asterisk indicates the significant level (***$p < 0.001$, **$p < 0.01$, *$p < 0.05$, $^{ns}p > 0.05$).

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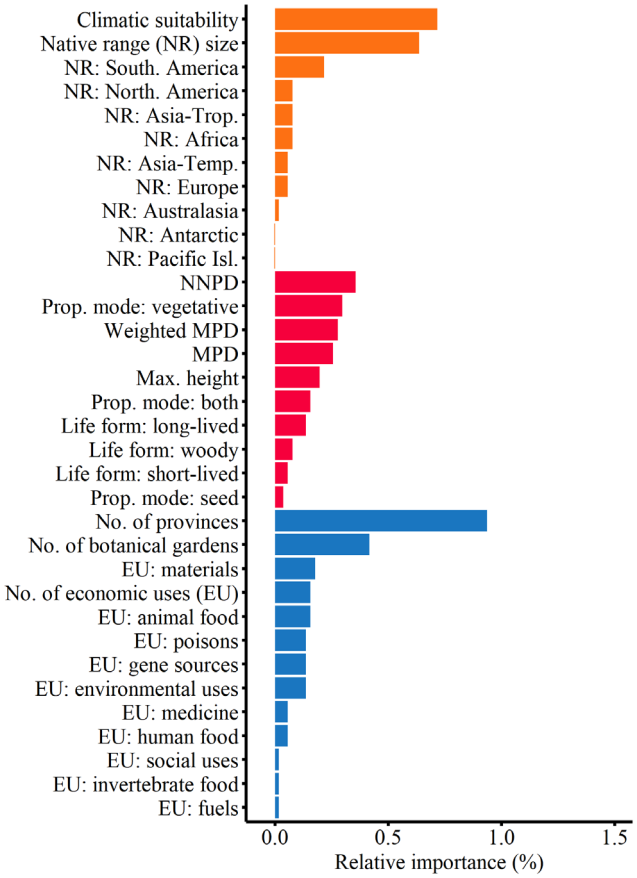

FIGURE 2 Relative importance for the 34 individual variables in the RF, based on the loss function (1 − AUC) after 1000 permutations. Within each variable group (orange: Environmental niches, red: Species traits, blue: Propagule pressure), the individual variables within each category were ordered according to a decrease in relative importance. EU, economic use; NR, native range.

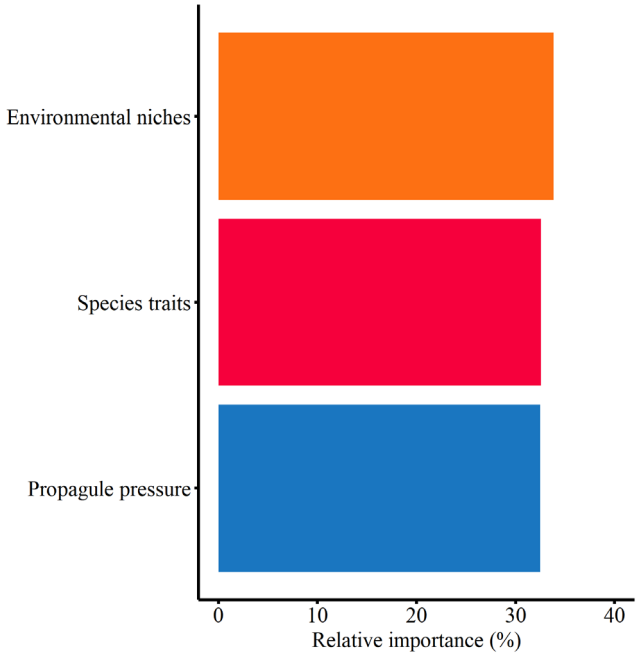

FIGURE 3 Relative importance of the three categories used in the RF, based on the loss function (1 − AUC) after 1000 permutations.

4 | DISCUSSION

4.1 | Single variables related to invasion success

Invasion success is strongly associated with characteristics related to species' environmental niches, traits, and propagule pressure (Dong, Yang, et al., 2024; Guo et al., 2024; Kinlock et al., 2022; Pyšek et al., 2015). For environmental niches, climatic suitability and native range size are key predictors, aligning with previous studies showing that many invasive species have large native ranges and largely retain their climatic niches in introduced ranges (Liu et al., 2020; Petitpierre et al., 2012). This is because taxa with larger native ranges typically exhibit broader niche breadths due to exposure to diverse abiotic and biotic conditions (Brown, 1984). A recent study found positive correlations between the range sizes of native and invasive European plants with their local abundance and habitat breadth (Fristoe et al., 2021). In addition, taxa with large native ranges often possess enhanced dispersal capabilities, such as wind- or animal-mediated dispersal, facilitating their spread in introduced ranges (Murray & Phillips, 2010; Tyler et al., 2015). Furthermore, taxa with large native ranges and high climatic suitability are more likely to have been introduced by humans for cultivation (Gravuer et al., 2008; Pyšek et al., 2004; van Kleunen et al., 2007). These findings underscore the importance of both climatic suitability and native range size as important predictors of invasion success.

Although species traits have been shown to facilitate plant invasions in numerous studies (e.g., Pyšek & Richardson, 2008; van Kleunen, Dawson, et al., 2010; Wang et al., 2017, 2024), their importance in predicting invasion was relatively low in our study. This may be because plant invasiveness involves a complex interplay of species traits, which can vary across different environmental contexts (Fristoe et al., 2021). For instance, invasive aquatic species like *Alternanthera philoxeroides* and *Eichhornia crassipes* exhibited high invasiveness in most of southern China due to their fast growth rate and clonal reproduction. However, their northern distribution is limited by cold winter temperatures, with *E. crassipes* being less cold-tolerant (Zhang et al., 2021). Additionally, *A. philoxeroides* faces range restrictions due to interspecific competition from *E. crassipes* and herbivory pressure from its biocontrol agent, *Agasicles hygrophila*, especially at lower latitudes (Zhang et al., 2021). These complex interactions highlight the difficulty of predicting invasion based solely on a single or few traits (Küster et al., 2008).

Our study demonstrated that alien plants with close phylogenetic relatedness to the native flora have a higher likelihood of becoming invasive, providing strong support for Darwin's pre-adaptation hypothesis. This finding aligns with a recent study by Qian (2023), which showed a higher phylogenetic relatedness to native flora

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among invasive plants in the provinces of China (Qian, 2023). However, contrasting results have emerged from studies in South Africa (Omer et al., 2022) and California (Strauss et al., 2006), where researchers have found either no significant relationship or a negative relationship between phylogenetic relatedness and invasion success. On the one hand, these discrepancies might reflect that some of the studies classified species as invasive based on the spread rate, while others did so based on the ecological or socioeconomic impact. Furthermore, the discrepancies might reflect that the effect of phylogenetic distance on invasion may change from positive to negative with latitude (Fan et al., 2023; Park et al., 2020). Irrespective of the exact reason, these conflicting findings from different studies highlight the complexity of the relationship between phylogenetic relatedness and invasiveness. They suggest that the influence of phylogenetic relatedness on the invasive potential of naturalized taxa might be context-dependent, varying with the native flora composition of different regions.

Propagule pressure is considered a critical factor in plant invasions (Colautti et al., 2006; Lockwood et al., 2005), and our study corroborated this widely accepted notion. Interestingly, we found that planting frequency, as indicated by the number of botanical gardens and provinces where planted, was nonlinearly correlated with invasion success. The initial positive relationship between planting frequency and invasion success likely stemmed from the frequent exchange of plant species among botanical gardens, both domestically and internationally. This practice increased the opportunities for alien plants to establish in new environments (Dong, Yang, et al., 2024; Kinlock et al., 2022; Maurel et al., 2016). The observed decline in invasion success at the higher planting frequencies could reflect the increased awareness about invasive species, leading many botanical gardens to remove the most invasive plants from their collections.

Our study also confirmed previous findings on global-scale plant naturalization (e.g., van Kleunen et al., 2020), showing that plants with one or more economic uses have a significantly higher likelihood of invasion compared with those without known economic use. Meanwhile, in China, we found that alien plants used for animal food, medicine, and as poisons were particularly prone to becoming invasive. This finding highlights a potential risk associated with the widespread use of herbal medicines in countries like China, as it may increase the chances of introducing and establishing potentially invasive plants (Weber et al., 2008; Yessoufou et al., 2021). Our data revealed that ~85% of invasive plants in China (e.g. *Datura stramonium* and *Phytolacca americana*) have a history of medicinal use. A similar trend was observed in South Africa, where ~28% of families containing alien plants (e.g. Euphorbiaceae) are extensively used in traditional medicine, potentially elevating their invasion risk (Yessoufou et al., 2021). The cultural and economic significance of alien plants in traditional medicines, animal feed, and chemicals (e.g. poisons) could lead to an increased reliance on imported alien plants in China (Weber et al., 2008). Consequently, the intentional introduction and cultivation of these plants for their valued uses inevitably increase propagule pressure and the risk of plant invasions (Guo et al., 2024).

4.2 | Grouped variables related to invasion success

Surprisingly, the 34 individual variables used in our model accounted for only a small proportion (6.17%) of the total predictive power of the RF. When grouped into three categories, these categories demonstrated significantly higher predictive power—13.5 to 17.7 times greater—than the cumulative relative importance of single variables within the same category. Among three predefined categories, the environmental niche category emerged as a slightly stronger predictor of invasion success, though differences between categories were not dramatic (Dong, Yang, et al., 2024). For invasive species management in China, strategies to prevent further invasions should not solely target specific plant traits and propagule pressure but should also consider the environmental conditions in both their native and introduced ranges. Moreover, the enhanced predictive power of grouped variables may stem from their ability to capture more comprehensive patterns and interactions within each category, which individual variables alone may not capture (Cutler et al., 2007; Küster et al., 2008). Notably, the role of grouped variables in predicting invasion has rarely been documented in ecological studies (see Cutler et al., 2007; Novoa et al., 2020). Therefore, we recommend category-based evaluation for a deeper understanding of how combined factors influence invasion success.

Additionally, our results from the randomization test provided clear evidence that the relative importance of the predefined category did not exceed the expected level when the number of individual variables within each category remained constant. This indicates that the relative importance of a category is predominantly driven by the number of variables it contains, rather than by specific properties or interactions of those variables. However, this finding does not undermine the validity of these three predefined categories. Instead, the non-significant results may imply that the contributions of individual variables are relatively evenly distributed across the categories, or that complex interactions among variables obscured category-level effects (Breiman, 2001). Overall, even when grouping variables may not directly improve model accuracy, it offers valuable insights into the collective influence of the variables on invasion success. This approach underscores the importance of considering both individual variables and their broader ecological contexts, facilitating a more comprehensive understanding of the factors driving plant invasions.

4.3 | Model refinement and future research directions

Our predefined categorization approach offers numerous advantages, including providing a framework to interpret complex models within ecologically meaningful contexts. It also facilitates comparisons with established invasion theories and serves as a foundation for more detailed data-driven analyses in future studies. However, to mediate a potentially large number of explanatory variables, we

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recommend exploring hybrid approaches that integrate ecological theory with data-driven variable groupings. For example, self-organizing maps (SOM) can serve as a powerful tool during the exploratory phase of feature selection, enabling the analysis of data structure and variable relationships (Kohonen, 2001). SOM's capacity to preserve topological relationships while reducing dimensionality is especially valuable for ecological datasets, where complex interactions among variables are prevalent. This capability offers promising avenues for future research, particularly in enhancing our understanding of invasive species patterns.

Due to our theory-informed predefined categories, we did not explore different combinations of variables across categories. For future studies, we recommend incorporating a flexible variable combination strategy to identify the most influential combinations of variables and to uncover potentially critical synergies that our current categorical method might overlook. For example, exploring combinations of multidimensional variables such as geographic range size, spread rate traits, and habitat breadth—that are particularly informative in recent invasion studies (e.g. Carboni et al., 2016; Catford et al., 2016; Fristoe et al., 2021)—could provide deeper insights into invasion dynamics.

5 | CONCLUSIONS

Our study provides valuable insights into the factors driving the invasion success of naturalized cultivated plants in China. By identifying key individual variables, we offer a scientifically grounded basis for prioritizing the surveillance of alien plant species with extensive native ranges and high climatic suitability. By demonstrating the enhanced interpretability of grouped variables, we highlight how the use of interpretable ML approaches can provide a deeper understanding of the complex interactions among individual variables within each predefined category. This comprehensive perspective can not only identify the most influential predictors of invasion success but also equip policymakers with evidence-based strategies for surveillance, early detection, and targeted intervention of invasive plants. In addition, with the future availability of biotic variables that capture species interactions (e.g. the presence or absence of natural enemies or pollinators), subsequent studies can achieve a more comprehensive understanding of invasion processes, improved predictive models, and more effective management strategies for invasive species both in China and globally. Overall, integrating interpretable ML techniques into national biosecurity frameworks will become essential for safeguarding biodiversity, maintaining ecosystem services, and protecting economic interests.

AUTHOR CONTRIBUTIONS

Bi-Cheng Dong, Fei-Hai Yu, and Mark van Kleunen conceived and designed the research; Bi-Cheng Dong, Ran Dong, Nicole L. Kinlock, and Mark van Kleunen contributed data; Bi-Cheng Dong and Ran Dong analysed the data with the support of Qiang Yang; Bi-Cheng Dong and Ran Dong drafted the article; and all authors revised the subsequent drafts. All authors contributed critically to revisions of the draft and gave final approval for publication.

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CONFLICT OF INTEREST STATEMENT

We declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data available via the Dryad Digital Repository https://doi.org/10.5061/dryad.2ngf1vj08 (Dong, Dong, et al., 2024). Additional Supporting Information will also be found in the online version of the article at the publisher's website.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Appendix S1: Phylogenetic relatedness and plant invasions.

Appendix S2: Methods.

Appendix S3: Random forest as a robust tool to predict invasion.

Appendix S4: Tables and Figures.

Appendix S5: Sources used to compile data on species traits.

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